

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	Program	Master of Computer Applications (MC305)
Master Subject Coordinator Name:	Mandeep Singh	Master Subject Coordinator E-Code:	E8730
Course Name	PL/SQL Lab	Course Code	25CAP-602

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
0	0	4	0	2.00	P

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Practical Examination (PRAC)	Practical (PRAC)

Mission of the Department	M1. To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2. To provide a framework through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3. To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4. To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5. To undertake societal activities for upliftment of rural/deprived sections of the society.
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)

PEO1	Establish a well-fortified computing foundation of successful professionals by applying computing fundamentals and domain-specific knowledge, demonstrating their innovative skills and considering social and environmental concerns.
PEO2	Undertake successful implementation of ethical solutions as an individual or a member or a leader of a team by investigating, analyzing, formulating and solving complex computing problems in multidisciplinary approaches using modern tools.
PEO3	Enhance professionalism and ethical attitude in the profession while communicating with local, national, and foreign peers, bound within regulations and leading to lifelong learning.
PEO4	Promote awareness for uplifting health, safety, legal, environmental, ethical and cultural diversity issues for serving the society.

Program Specific OutComes(PSOs)

PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)

PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Fundamentals of Database Systems	Elmasri & Navathe	Fourth Edition	Pearson	2004
2	Introduction to Database Management System	C. J. Date	Eighth Edition	Addison-Wesley	2003
3	Database Management Systems	Raghu Ramakrishnan and Johannes Gehrke	Third Edition	McGraw-Hill Education	2002
4	Database System Concepts	Abraham Silberschatz, Henry F. Korth, and S. Suda	Seventh Edition	McGraw-Hill Education	2019

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	An Introduction to Database Management Systems	Bipin C. Desai	Revised Edition	Galgotia Publications Pvt Ltd	2013
2	Principles of distributed database	M. Tamer Ozsu	Forth edition	Springer	2019

Course OutCome	
SrNo	OutCome
CO1	Understand, analyze and apply common SQL statements including DDL, DML and DCL statements to perform different operations.
CO2	Design different views of tables for different users and to apply embedded and nested queries.
CO3	Design and implement a database for a given problem according to well known design principles that balance data retrieval performance with data consistency.
CO4	Demonstrate and understand relational algebra in Database which is helpful to design related database software components.
CO5	Understand, analyze and apply common SQL statements including DDL, DML and DCL statements to perform different operations.

Lecture Plan Preview-Practical

Unit No	ExperimentNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer(s)
1	1	Horizontal Fragmentation and Relational Algebra Optimization in a Distributed Database Environment	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO1,CO2
1	2	Query Optimization Using Selection and Projection Pushdown in Relational Algebra	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO1,CO2
2	3	Amber's conglomerate corporation just acquired some new companies. Each of the companies follows this hierarchy	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO2,CO3
2	4	Multi-Table Join, Aggregation, and Conditional Output Filtering Using SQL	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO3,CO4
3	5	Query the two cities in STATION with the shortest and longest CITY names, as well as their respective lengths (i.e.: number of characters in the name). If there is more than one smallest or largest city, choose the one that comes first when ordered alphabetically.	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO4

3	6	Write a query to output the start and end dates of projects listed by the number of days it took to complete the project in ascending order. If there is more than one project that have the same number of completion days, then order by the start date of the project.	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO4,CO5
3	7	Project 1 Title: Online Bookstore Management System	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO4,CO5
3	8	Project 2 Title: Employee Management System	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO4,CO5
3	9	Project 3:Distributed Database for Multi-branch Company	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO4,CO5
3	10	Project 4:Social Network Analysis System	,T- Database System Concepts,T- Fundamentals of Database Sys,T- Introduction to Database Man,T-Database Management Systems,R- An Introduction to Database ,R-Principles of distributed data	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulation Practical,Video Demonstration	CO4,CO5

Assessment Model			
Sr No	Assessment Name	Exam Name	Max Marks
1	20PRAC01	External Viva / Voce	40
2	20PRAC01	Experiment-1	30
3	20PRAC01	Experiment-2	30
4	20PRAC01	Experiment-3	30
5	20PRAC01	Experiment-4	30

6	20PRAC01	Experiment-5	30
7	20PRAC01	Experiment-6	30
8	20PRAC01	Experiment-7	30
9	20PRAC01	Experiment-8	30
10	20PRAC01	Experiment-9	30
11	20PRAC01	Experiment-10	30
12	20PRAC01	Mid-Term Test	15

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2			1	1									
CO2	2			1	1									1
CO3	1	1		2	2									3
CO4	3		3	2	2									2
CO5	3		3	2	2									2
Target	2.2	1	3	1.6	1.6	0	0	0	0	0	0	0	0	2

